

WT Exp-6

➤ **Aim :-**

Interface the Arduino, connect the components together and run the project.

➤ **Code :-**

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>

#define SS_PIN 10
#define RST_PIN 9
#define BUZZER 2 // Buzzer pin
#define SERVO_PIN 6 // Servo pin

MFRC522 mfrc522(SS_PIN, RST_PIN); // RFID instance
Servo myServo; // Servo instance
LiquidCrystal_I2C lcd(0x27, 16, 2); // LCD instance with I2C address 0x27

void setup() {
  Serial.begin(9600);
  SPI.begin();
  mfrc522.PCD_Init();

  myServo.attach(SERVO_PIN);
  myServo.write(0); // Ensure servo starts at 0°

  pinMode(BUZZER, OUTPUT);
  noTone(BUZZER);

  lcd.init();
  lcd.clear();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print(" Put card on ");
  lcd.setCursor(0, 1);
  lcd.print(" reader ");
}

void loop() {
  if (!mfrc522.PICC_IsNewCardPresent()) {
```

```

    return;
}
if (!mfrc522.PICC_ReadCardSerial()) {
    return;
}

lcd.clear();
lcd.setCursor(0, 0);
lcd.print("UID tag:");

String content = "";
for (byte i = 0; i < mfrc522.uid.size; i++) {
    lcd.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
    lcd.print(mfrc522.uid.uidByte[i], HEX);
    content.concat(String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " "));
    content.concat(String(mfrc522.uid.uidByte[i], HEX));
}
content.toUpperCase();

lcd.setCursor(0, 1);
if (content.substring(1) == "53 37 5A E4") { // Replace with your card's UID
    lcd.print("Access Granted");
    tone(BUZZER, 500);
    delay(300);
    noTone(BUZZER);
    myServo.write(90); // Move servo to 90° (open position)
    delay(3000);      // Hold open for 3 seconds
    myServo.write(0); // Move servo back to 0° (closed position)
    delay(1000);      // Small delay before resuming loop
} else {
    lcd.print("Access Denied");
    tone(BUZZER, 300);
    delay(2000);
    noTone(BUZZER);
}
delay(2000);
}

```

➤ **Practical :-**

Connecting the Arduino with all of its components :

RFID Reader to Arduino (SPI Protocol)

RFID Reader Pin	Arduino Pin	Description
SDA	10	Slave Select (SS)
SCK	13	Serial Clock (SCK)
MOSI	11	Master Out Slave In (MOSI)
MISO	12	Master In Slave Out (MISO)
GND	GND	Ground
3.3V	3.3V	Power Supply

LCD Display to Arduino (I2C Protocol)

LCD Display Pin	Arduino Pin	Description
SCL	A5	Serial Clock Line (SCL)
SDA	A4	Serial Data Line (SDA)
VCC	5V	Power Supply
GND	GND	Ground

Servo Motor to Arduino

Servo Motor Pin	Arduino Pin	Description
Control Pin	D7	Signal Control
Power	5V	Power Supply
Ground	GND	Ground

Buzzer to Arduino

Buzzer Pin	Arduino Pin	Description
+ (Positive)	D2	Signal Input
- (Negative)	GND	Ground

➤ Circuit Diagram :-

